

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M.SANDHYA(192312421) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWERS

1.Solution

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints:

1. D1 $\neq$ Night

2. D2 must be before D3 ($M < A < N$)
3. Only one doctor per shift
4. $D3 \neq \text{Morning}$
5. Each doctor gets exactly one shift

Step 1: Assign D1

Possible: M, A (not N)

Try **D1 = Morning**

Step 2: Assign D2

Remaining: A, N

Try **D2 = Afternoon**

Step 3: Assign D3

Remaining: N

Try **D3 = Night**

Constraint Check

- $D1 \neq \text{Night} \rightarrow (\text{Morning})$
- $D3 \neq \text{Morning} \rightarrow (\text{Night})$
- $D2 \text{ before } D3 \rightarrow \text{Afternoon} < \text{Night}$
- One doctor per shift
- All assigned

Final Valid Schedule

D1 $\rightarrow$ Morning

D2 $\rightarrow$ Afternoon

D3 $\rightarrow$ Night

2. Solution

Grid Path Finding using A* (Manhattan Distance Heuristic)

Given:

- Grid size: 5×5
- Start: $S = (0,0)$
- Goal: $G = (4,4)$
- Obstacles: $(1,1), (1,2), (2,2), (3,1)$
- Moves: Up, Down, Left, Right
- Cost per move = 1

Heuristic Function (Manhattan Distance)

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

A* Evaluation Function

$$f(n) = g(n) + h(n)$$

$g(n)$ = cost from start

$h(n)$ = estimated cost to goal

Step-by-Step Path Selection (A*)

Start at **(0,0)**

1. $(0,0) \rightarrow f = 0 + 8 = 8$
2. Move to $(0,1) \rightarrow f = 1 + 7 = 8$
3. Move to $(0,2) \rightarrow f = 2 + 6 = 8$
4. Move to $(0,3) \rightarrow f = 3 + 5 = 8$
5. Move to $(0,4) \rightarrow f = 4 + 4 = 8$
6. Move to $(1,4) \rightarrow f = 5 + 3 = 8$
7. Move to $(2,4) \rightarrow f = 6 + 2 = 8$
8. Move to $(3,4) \rightarrow f = 7 + 1 = 8$
9. Move to $(4,4) \rightarrow f = 8 + 0 = 8$

Final Path

$(0,0) \rightarrow (0,1) \rightarrow (0,2) \rightarrow (0,3) \rightarrow (0,4) \rightarrow (1,4) \rightarrow (2,4) \rightarrow (3,4) \rightarrow (4,4)$

Total Cost

= 8 moves

3. Solution

a) Analysis of Constraints

1. **Movement (Up, Down, Left, Right)**
→ The robot explores the grid step-by-step. No diagonal shortcuts, so path length may increase.
2. **Obstacles (Blocked Cells)**
→ The robot must avoid these cells. It may need to backtrack and find alternative paths.
3. **Limited Sensing (Adjacent Cells Only)**
→ The robot does not know the full map initially. It must explore and update knowledge while moving (partial observability).
4. **Movement Cost = 1**
→ Every step adds cost, so shorter paths are preferred.
5. **Risky Zones (Extra Cost = +2)**
→ Entering risky cells increases total cost (total = 3). The robot should avoid them unless necessary.
6. **Minimize Total Cost**
→ The robot must choose the safest and cheapest path, not just the shortest.
7. **Dynamic Knowledge (Online Search)**
→ The robot updates its map while moving and makes decisions in real-time.

b) Solution Approach (Online Uniform Cost Search)

Chosen Method: Online Search Agent using Uniform Cost Search (UCS)

Steps:

1. Initialize

1. Start at position S
2. Known map = empty
3. Cost $g(n) = 0$

2. Sense Environment

1. Observe adjacent cells (up, down, left, right)
2. Mark cells as free, blocked, or risky

3. Expand Possible Moves

1. Generate valid neighboring cells
2. Ignore blocked cells

4. Assign Cost

1. Normal cell = +1
2. Risky cell = +3

5. Select Next Move

1. Choose the path with **minimum cumulative cost (UCS principle)**

6. Move and Update

1. Move to selected cell
2. Update map with new observations

7. Repeat

1. Continue until Goal (G) is reached

c) AI Concepts Applied

1. Intelligent Agent

- The robot acts as an **agent**:
 - **Percepts:** Adjacent cells (free, blocked, risky)
 - **Actions:** Move in 4 directions
 - **Goal:** Reach survivor safely

2. Rationality

The robot is **rational** because:

- It chooses actions that minimize total cost
- Avoids risky areas unless necessary
- Makes best decision based on available information

3. Environment Type

- **Partially Observable:** Cannot see entire grid
- **Deterministic:** Movement outcomes are predictable
- **Sequential:** Current action affects future steps
- **Dynamic:** Knowledge changes as robot explores
- **Discrete:** Grid-based environment

d) Justification of Approach

- **Uniform Cost Search (UCS)** ensures the **minimum-cost path**, considering both normal and risky cells.
- **Online Search** is necessary because:
 - The robot does not know the full environment initially
 - It must learn and adapt while moving
- This combination:
 - Handles **partial observability**
 - Minimizes cost effectively
 - Ensures safe navigation

Therefore, **Online UCS** is the most suitable approach for this problem.

Final Conclusion

The rescue robot uses an **online, cost-based search strategy** to dynamically explore the environment, avoid obstacles, minimize risk, and efficiently reach the survivor.